

SMART INDIA HACKATHON 2026

TrainETA — SIH PPT Content Pack

Project: Indian Railway Dynamic ETA Prediction System

Problem Statement ID: SIH26028

Problem Statement: Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

Purpose: Ready-to-use content for the PPT team member. The official SIH template allows a maximum of six slides including the title slide and recommends concise points, diagrams, infographics and pictures.

SLIDE 1 — TITLE

SMART INDIA HACKATHON 2026

Problem Statement ID: SIH26028

Problem Statement: Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

Project Title: Indian Railway Dynamic ETA Prediction System

Theme: Transportation & Logistics (confirm exact portal wording)

PS Category: Software

Team ID: [Your Team ID]

Team Name: [Registered Team Name]

Tagline: Real-time train tracking + historical delay intelligence + ML-based dynamic ETA prediction

Visual: Simple railway/live-tracking visual; keep the title slide uncluttered.

SLIDE 2 — IDEA / PROPOSED SOLUTION

Our Solution

- Live train position & speed
- Current and next station
- Current delay
- Historical train delay patterns
- Machine Learning-based ETA prediction
- Scheduled railway route & distance
- Operational factors
- Live route visualization

How it works

**Live Train Data + Historical Delay Data + Timetable/Route + Operational Factors → ETA Prediction Engine
→ Dynamic ETA Forecast → Station-wise Arrival Predictions → Live Railway Route Map**

Problem Addressed

- Static timetable ETA cannot reflect changing delays.
- Train delays can propagate across subsequent stations.
- Actual train movement differs from scheduled movement.
- Passengers need updated arrival predictions, not only scheduled timings.

Innovation / Uniqueness

- Live + historical + ML hybrid approach
- Dynamic prediction instead of timetable-only ETA
- Station-wise future ETA forecasting
- Live train position displayed on railway route
- Operational factors incorporated into prediction
- Backend designed to integrate real-time railway data

Visual: Use a simple pipeline diagram rather than paragraphs.

SLIDE 3 — TECHNICAL APPROACH

Technology Stack

Layer	Technologies
Frontend	Next.js, React, TypeScript
UI	Tailwind CSS
Backend	Python, FastAPI
Database	PostgreSQL
ML	XGBoost, Python
Live Data	RailRadar API
Maps	MapLibre GL, OpenStreetMap/OpenFreeMap
API	REST APIs
Development	Git, GitHub, VS Code
Deployment	Render

Prediction Pipeline

Railway Timetable + Historical Delay Dataset + RailRadar Live Data → Data / ETA Prediction Engine → Current ETA + Station-wise Forecast + Delay Analysis → Next.js Dashboard

ML / ETA Method: Historical data → Feature engineering → XGBoost prediction → Live-condition adjustment → Station-wise ETA

Key live inputs: Current delay, current speed, train position, remaining distance, route progress, scheduled timings, historical delay behaviour and operational conditions.

Working Prototype: Insert the actual TrainETA dashboard/map screenshot. Current prototype demonstrates train 12919 with live position, 69% completed, 502.2 km remaining and live speed of 97.3 km/h in the captured screen.

Important accuracy wording: Do not claim 99%/100% accuracy.

SLIDE 4 — FEASIBILITY & VIABILITY

Feasibility

- Technical: FastAPI REST APIs, PostgreSQL, XGBoost inference and MapLibre support a modular architecture.
- Data: Historical railway delay data and timetable/route information are available; live train data is integrated through an API.
- Deployment: Frontend/backend separation supports cloud deployment and future data-source integration.

Challenges → Solutions

Challenge	Approach
Live train position availability	Integrate live railway API
Historical vs live data mismatch	Route/station alignment
Delay propagation	Historical + ML prediction
Changing operational conditions	Dynamic operational adjustments
Intermediate/pass-through stations	Route alignment and distance-based fallback
API/data availability	Modular data-source architecture
ML leakage risk	Chronological offline evaluation

Scalability: More historical data → better feature engineering → model retraining → improved ETA predictions → more trains/routes/stations.

SLIDE 5 — IMPACT & BENEFITS

Target Users

- Passengers
- Railway Operations
- Station Authorities
- Control Rooms
- Travel / Railway Applications

Benefits

Area	Benefits
Passengers	More realistic arrival information; better journey planning; reduced uncertainty during delays; station-wise ETA visibility
Railway Operations	Dynamic visibility of train movement; delay propagation insights; operational monitoring support; data-driven decision making
Economic	Better utilization of passenger/station resources; reduced uncertainty around delayed train arrivals; potential supply chain optimization
Social	Improved passenger information; better access to real-time train status; reduced dependence on static timetable
Environmental	Primarily a digital/information-based solution; no additional physical infrastructure required for the software system

Impact statement: Transforms static railway schedules into dynamic, data-driven arrival predictions.

SLIDE 6 — RESEARCH & REFERENCES

Research / Data Sources

- Smart India Hackathon 2026 — Problem Statement SIH26028: Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains.
- Indian Railway historical route & delay dataset — used for historical delay analysis and model development.
- RailRadar — live train-data source used in the working prototype.
- XGBoost — machine-learning framework used for ETA prediction.
- FastAPI Documentation — backend REST API implementation.
- PostgreSQL Documentation — railway data storage and retrieval.
- MapLibre / OpenStreetMap / OpenFreeMap — railway route visualization.

Experimental Evidence

Evaluation journeys	57,465
MAE	25.0 minutes
RMSE	67.8 minutes
Mean error	+1.3 minutes
Historical component vs current-delay-only	8.9% MAE improvement

Accuracy / implementation caveats for the PPT team

- Do not claim 99% or 100% prediction accuracy.
- Do not describe RailRadar as a direct Indian Railways live API.
- Do not claim actual preceding-train detection; the current implementation uses a congestion-based operational proxy.
- Present the ML model as the configured ETA model and clearly distinguish prototype/live integration from future production enhancements.

Official template requirements: maximum 6 slides including title; use points/diagrams/infographics/pictures; keep explanations precise and easy to understand; submit the final file as PDF.